Probabilistic College Basketball Ratings: A Ridge Prior Is Worth Three Weeks of Data

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Abstract

We develop a probabilistic framework for college basketball team ratings that produces a full posterior distribution over team quality, rather than a point estimate. Two models are derived and compared: a multiplicative fixed-point system modeled after KenPom (Model 1) and a ridge regression model with an exact Gaussian posterior (Model 2). Both are validated under a strict *one-step-ahead* protocol on the 2025–26 NCAA season: for each week, the model is fit on all prior games and evaluated on that week's games exclusively. Model 2 achieves a season-wide RMSE of 14.14 pts/100 possessions versus 15.10 for Model 1. The gap is largest early in the season, where Model 1 is nearly underdetermined: in week 1, Model 1 RMSE exceeds 22 while Model 2 opens at 14.6 — empirically confirming that the ridge prior provides the equivalent of approximately three weeks of additional data. Model 2's posterior predictive intervals are empirically well-calibrated: a 90% interval contains roughly 90% of actual outcomes across all coverage levels. The small systematic negative bias observed in both models is a direct consequence of within-season team improvement, pointing naturally at a dynamic-coefficient extension.

1 Introduction

College basketball rating systems — KenPom [Pomeroy, 2024], BartTorvik, ESPN's BPI — universally report point estimates of team quality. A team is assigned an adjusted offensive efficiency, a defensive efficiency, and a tempo, and these numbers are used for ranking, seeding, and matchup analysis.

This approach has a structural limitation: point estimates give no indication of how confident the model is. A team that has played three games has the same representation as a team that has played thirty. Early in the season, when teams have played five to ten games each and the schedule graph is sparse, the estimates are largely determined by prior assumptions that are never made explicit.

We make those assumptions explicit. Specifically, we show that:

1. A Gaussian prior over team effects (the implicit assumption of ridge regression) produces well-posed inference from the first game of the season, with quantifiable uncertainty.

2. This prior is worth approximately three weeks of additional data in early-season forecast accuracy, relative to the fixed-point system that has no explicit regularization.

3. The resulting posterior predictive intervals are empirically well-calibrated across all coverage levels, enabling honest probabilistic statements about matchup outcomes.

4. The systematic underprediction bias observed in both models is interpretable as a within-season team improvement effect, and points directly at a state-space extension.

The framework is built on a common data structure — offensive efficiency per 100 possessions, one observation per team per game — and produces a KenPom-compatible summary (AdjO, AdjD, AdjPace) as a *derived* quantity from the posterior, rather than a primitive estimate.

2 Data

All data are sourced from RealGM via a custom scraper into a local SQLite database. The 2025–26 NCAA Division I season contains 6,305 unique games involving approximately 350 teams. Each game yields two observations (one per team side), for a total of 12,610 rows.

The primary outcome variable is offensive efficiency:

$$e_{\text{off},ig} = \frac{\text{pts}_{ig}}{\text{poss}_{ig}} \times 100 \tag{1}$$

where pts_{ig} is points scored and poss_{ig} is possessions used by team i in game g. Possessions are estimated from boxscore data using the standard formula. Observations with fewer than 25 possessions (garbage time, technical fouls) are excluded.

For the 2025–26 season, the distribution of e_{off} has mean ≈ 107 pts/100 and standard deviation ≈ 16 pts/100, with a near-Gaussian shape and heavy right tail from blowout victories.

3 Models

3.1 Model 1: Fixed-Point Iteration

Model 1 follows the KenPom methodology [Pomeroy, 2024]. Each team i is assigned latent scalars $(O_i, D_i, R_i) \in \mathbb{R}^3$ representing adjusted offensive efficiency, adjusted defensive efficiency, and adjusted pace.

Update equations. Let $w_g = p_g/\bar{p}$ be possession weights and $L(h) \in \{1.014, 1.000, 0.986\}$ a home/neutral/away location factor. The self-consistency conditions are:

$$O_i^{(t+1)} = \frac{\sum_g w_g \cdot e_{\text{off},ig} \cdot (\bar{\mu}/D_j^{(t)}) \cdot L(h)^{-1}}{\sum_g w_g} \tag{2}$$

$$D_i^{(t+1)} = \frac{\sum_g w_g \cdot e_{\text{off},jg} \cdot (\bar{\mu}/O_j^{(t)}) \cdot L(h)}{\sum_g w_g} \tag{3}$$

where the sums are over games involving team i, and j denotes the opposing team. Each iteration renormalizes so that $\mathbb{E}_i[O_i] = \mathbb{E}_i[D_i] = \bar{\mu}$. Convergence is declared when $\max_i |O_i^{(t+1)} - O_i^{(t)}| < 10^{-6}$.

Prediction. Given converged estimates, the predicted offensive efficiency for team i against opponent j at location h is:

$$\hat{e}_{\text{off}} = O_i \cdot \frac{D_j}{\bar{\mu}} \cdot L(h) \tag{4}$$

Uncertainty. Model 1 has no analytic posterior. We use a game-level parametric bootstrap: resample games (with both rows of each game moving together to preserve within-game correlation), refit the fixed-point iteration, and collect the resulting (O_i, D_i, R_i) as posterior draws.

Limitation. Model 1 has no regularization. With $T \approx 350$ teams and a sparse early-season schedule graph, the system of equations in (2)–(3) is nearly underdetermined. Teams with few games receive extreme estimates that are subsequently corrected only as more games are played. This instability is the core motivation for Model 2.

3.2 Model 2: Ridge Regression with Exact Gaussian Posterior

Observation model. The additive generative model for offensive efficiency is:

$$e_{\text{off},ig} = \mu + o_i - d_j + \eta \cdot h_g + \varepsilon_g, \qquad \varepsilon_g \sim \mathcal{N}\left(0,\ \frac{\sigma^2}{p_g}\right) \tag{5}$$

where μ is the league mean, $o_i \in \mathbb{R}$ is team i's offensive coefficient, $d_j \in \mathbb{R}$ is team j's defensive coefficient (higher = better defense), $h_g \in \{-1, 0, +1\}$ encodes home/neutral/away for team i, and η is the home-court advantage. Observations are weighted by possessions p_g to downweight short games.

Prior. We place independent Gaussian priors on team effects:

$$o_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \lambda^{-1}), \qquad d_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \lambda^{-1}) \tag{6}$$

with $\lambda = 1.0$. No prior is placed on μ or η (i.e. $\lambda_\mu = \lambda_\eta = 10^{-4}$, near-flat).

Posterior. Let $\theta = (\mu, o_1, \ldots, o_T, d_1, \ldots, d_T, \eta)^\top \in \mathbb{R}^{2T+2}$ and assemble the design matrix $X \in \mathbb{R}^{N \times (2T+2)}$ with rows encoding (5). Let $W = \text{diag}(p_1/\bar{p}, \ldots, p_N/\bar{p})$ and $\Lambda = \text{diag}(\lambda_\mu, \lambda, \ldots, \lambda, \lambda_\eta)$. The MAP estimate is:

$$\hat{\theta} = \left(X^\top W X + \Lambda\right)^{-1} X^\top W y \tag{7}$$

Since the prior is Gaussian and the likelihood is Gaussian, the exact posterior is:

$$p(\theta \mid \mathcal{D}) = \mathcal{N}\left(\hat{\theta},\ \hat{\sigma}^2 \left(X^\top W X + \Lambda\right)^{-1}\right) \tag{8}$$

where $\hat{\sigma}^2 = (y - X\hat{\theta})^\top W (y - X\hat{\theta})/(N - 2T - 2)$. Posterior samples are drawn via Cholesky decomposition of the covariance.

KenPom summary. The three-number summary is derived from the posterior:

$$\text{AdjO}_i = \mu + o_i, \qquad \text{AdjD}_i = \mu - d_i, \qquad \text{AdjPace}_i = \exp(\gamma_0 + r_i) \tag{9}$$

where $(\gamma_0, r_1, \ldots, r_T)$ are fit by a separate log-linear pace model. Crucially, these are *derived* posterior summaries, not primitives: each draw from the posterior yields a full set of $(O_i, D_i, \text{Pace}_i)$ values, so uncertainty propagates through to the KenPom numbers.

Pace model. Pace is modeled log-linearly with one row per game:

$$\log \text{poss}_g = \gamma_0 + r_i + r_j + \gamma_h \cdot |h_g| + \varepsilon_g^{(r)} \tag{10}$$

where both teams receive a +1 in their respective r columns.

4 Evaluation Protocol

One-step-ahead (OSA) forecasting. The evaluation protocol is designed to strictly mirror deployment conditions. Let $\mathcal{G}_t = \{g : \text{date}(g) < t\}$ denote all games played before date t. For each week t in the season:

1. Fit the model on $\mathcal{G}_t$
2. Predict efficiency for each game g with $t \leq \text{date}(g) < t + 7$
3. Record residuals $r_{ig} = \hat{e}_{\text{off},ig} - e_{\text{off},ig}$

We require $|\mathcal{G}_t| \geq 300$ before the first prediction window, which begins the second week of November.

This protocol is stricter than the common 80/20 train/test split: the model never touches any game before predicting it. The 80/20 split is used only for calibration analysis in Section 5.3.

Metrics. For each week t we report:

$$\text{RMSE}_t = \sqrt{\frac{1}{N_t} \sum_{g \in W_t} \sum_i r_{ig}^2}$$

$$\text{MAE}_t = \frac{1}{N_t} \sum_{g \in W_t} \sum_i |r_{ig}|$$

$$\text{bias}_t = \frac{1}{N_t} \sum_{g \in W_t} \sum_i r_{ig}$$

where W_t is the set of observations in week t. Cumulative versions aggregate all residuals through week t.

Table 1: One-step-ahead RMSE (pts/100 poss) for selected weeks, 2025–26 season.

Week	Training games	Model 1	Model 2
Nov 14	676	22.90	14.63
Nov 21	1,032	17.83	15.80
Nov 28	1,425	15.14	14.29
Dec 05	1,748	14.31	13.94
Jan 02	2,797	14.75	14.28
Feb 06	4,503	13.79	13.68
Mar 13	6,147	12.35	12.27
Season	6,305	**15.10**	**14.14**

5 Results

5.1 Early-Season Instability of the Fixed-Point System

Table 1 summarizes RMSE for selected weeks. In week 1 (mid-November, $|\mathcal{G}_t| \approx 700$ games across 350 teams), Model 1 achieves RMSE = 22.9 while Model 2 opens at 14.6.

Model 1 does not stabilize (RMSE < 15.5) until approximately December 5th — roughly three weeks into the season. During those three weeks, Model 2 is already operating near its season-average accuracy. We interpret this as the empirical value of the ridge prior: it encodes the same information that three weeks of games provide to Model 1.

By mid-January, the gap narrows to ≈ 0.1–0.2 pts/100. The cumulative season RMSE gap is 0.96 pts/100 (14.14 vs. 15.10), dominated by the early-season disparity.

Proposition 1 (Informal)**.** *The ridge prior in Model 2 is equivalent in forecast value to approximately three additional weeks of game data for Model 1 in early-season prediction.*

This is not a theoretical claim but an empirical observation: Figure **??** (see notebook) shows cumulative RMSE curves converging, with Model 2's initial level matching Model 1's level at the point of stabilization $\approx$ December 5th.

5.2 Season Summary Statistics

Table 2 gives season-wide metrics across 10,562 test observations.

Table 2: Season-wide one-step-ahead forecast metrics, 2025–26.

Model	RMSE	MAE	Bias
Model 1 (fixed-point)	15.097	11.909	−1.449
Model 2 (ridge)	14.138	11.244	−1.175

5.3 Calibration of Posterior Predictive Intervals

To assess calibration, we fit Model 2 on the first 80% of games (by game identifier, approximately chronological) and construct posterior predictive intervals for the held-out 20%.

The predictive distribution for a single observation is:

$$p(\tilde{e}_{ij} \mid \mathcal{D}) \approx \frac{1}{S} \sum_{s=1}^{S} \mathcal{N}\Big(\hat{e}_{ij}^{(s)},\ \sigma_{\text{eff}}^2\Big), \qquad \theta^{(s)} \sim p(\theta \mid \mathcal{D}) \tag{11}$$

where $\sigma_{\text{eff}}^2 = \mathbb{E}[(\hat{e} - e_{\text{off}})^2]$ is estimated from training residuals.

The calibration curve (Figure **??**) plots empirical coverage against nominal coverage for 60 levels from 2% to 99%. Reference values:

Nominal	Empirical
50%	49%
80%	80%
90%	88%
95%	93%

The model is *very slightly underconfident* at high coverage levels (95% nominal → 93% empirical), meaning the intervals are marginally wider than necessary. This is the safer failure mode: overconfident intervals (too narrow) miss actual outcomes and mislead downstream decisions.

5.4 The Bias Signal

Both models exhibit a consistent negative bias of ≈ -1.2 to -1.5 pts/100 (predictions are below actuals). This pattern is not attributable to model misspecification in the standard sense.

The model assumes o_i is constant within the season. In reality, team offenses improve as players gain experience, coaching adjustments take hold, and injured players return. A model trained on November and December games and asked to predict March games will therefore systematically under-predict, because it has learned an earlier, weaker version of each team's offense.

This interpretation generates a testable prediction: the bias should be larger for games later in the season. We observe this in Figure **??** (notebook): the weekly bias is near zero in early December and becomes consistently negative by late January, growing in magnitude through the tournament.

The natural fix is to replace the static o_i with a dynamic state:

$$o_i(t) = o_i(t-1) + \xi_t, \qquad \xi_t \sim \mathcal{N}(0, \tau^2) \tag{12}$$

yielding a Kalman-filter-style inference problem. We leave this extension for future work.

6 Team Rating Trajectories and Uncertainty

Re-fitting Model 2 weekly on cumulative data produces team net-rating trajectories $\hat{o}_i(t) + \hat{d}_i(t)$ with posterior $\pm 1\sigma$ and $\pm 2\sigma$ bands at each time point.

Several observations follow from the 2025–26 season:

- Early-season ($|\mathcal{G}| < 1500$): posterior σ for a typical top-25 team is 1.5–3.0 pts/100. At this scale, most teams' credible intervals overlap substantially.
- By mid-January ($|\mathcal{G}| \approx 3800$): σ has collapsed to 0.3–0.4 pts/100 for teams with 15+ games. The hierarchy is clear.
- The rate of uncertainty reduction is approximately uniform across teams at similar stages of the schedule, confirming that variance is driven by sample size rather than structural instability.
- Elite teams (Michigan, Duke, Arizona in 2025–26) achieve smaller late-season σ than mid-major teams, consistent with playing more high-information (competitive) games.

7 Pre-Game Predictive Distributions

For any forthcoming game between teams i and j, the full predictive distribution (11) is available for both teams. The overlap between $p(\tilde{e}_i \mid \mathcal{D})$ and $p(\tilde{e}_j \mid \mathcal{D})$ provides a visual and quantitative measure of competitiveness.

For six top-25 matchups sampled across the 2025–26 season, we observe:

- Games between teams with similar net ratings produce nearly identical predictive densities, reflecting genuine uncertainty about outcomes.
- Games involving a large net-rating gap show clearly separated distributions, with actual outcomes (dotted lines) falling within the bulk of the predicted density in all six cases — a direct visual calibration check.
- The width of individual densities ($\approx$ 25–30 pts/100 at the 95% PI) is driven primarily by $\sigma_{\text{eff}} \approx 14$ pts/100 (irreducible game randomness), with parameter uncertainty contributing $\approx$ 1–2 pts/100.

8 Discussion

8.1 Ratings as Posterior Summaries

The central conceptual contribution of this work is framing team ratings as *derived summaries of a posterior distribution*, rather than primitives. Formally:

$$\theta \sim p(\theta \mid \mathcal{D}) \quad \Longrightarrow \quad S(\theta) = (\text{AdjO}_i, \text{AdjD}_i, \text{AdjPace}_i)_{i=1}^{T} \tag{13}$$

This means that every downstream quantity — KenPom-style ratings, matchup predictions, win probabilities — inherits calibrated uncertainty from the same generative object. The framework subsumes the point-estimate case as the posterior mean, while providing uncertainty estimates that are provably calibrated.

8.2 Comparison to Existing Work

Bayesian approaches to sports ratings have a long history [Glickman and Stern, 1998, Kovalchik, 2016]. The closest methodological relative is the RAPM (Regularized Adjusted Plus-Minus) framework from basketball analytics [Sill, 2010], which uses ridge regression on player-level data. Our contribution extends this to team-level ratings with an explicit posterior and a rigorous evaluation protocol.

The key novelty is the one-step-ahead validation, which is largely absent in the sports analytics literature. Most published evaluations use either a single train/test split or report in-sample fit, neither of which reflects deployment conditions.

8.3 Limitations and Extensions

Static coefficients. The primary limitation is the assumption of time-invariant o_i and d_i. The consistent negative bias (Section 5.4) is a direct consequence. The Kalman filter extension in (12) is the natural remedy.

Schedule strength circularity. Adjusted efficiency is computed relative to opponents' adjusted efficiency, creating a circular dependency that the fixed-point iteration resolves for Model 1 and the ridge regression approximates for Model 2 (via the $o_i - d_j$ encoding). True opponent-adjusted inference would require a hierarchical model.

Player-level structure. Team-level ratings aggregate over lineup combinations. The next modeling step is a player-level decomposition where team efficiency is a function of the active lineup, enabling transfer of information across teams sharing personnel. This connects naturally to the graph neural network architecture of Danielson [2024].

9 Conclusion

We have demonstrated that a ridge prior over team effects in a Gaussian observation model produces:

1. Early-season forecast accuracy equivalent to three additional weeks of game data compared to the standard fixed-point approach.

2. Season-wide RMSE of 14.14 pts/100 versus 15.10 for the fixed-point system, under strict one-step-ahead evaluation.

3. Empirically near-perfect calibration across all nominal coverage levels, with slight and benign underconfidence at high levels.

4. Interpretable systematic bias as a within-season team improvement signal, pointing to a dynamic-coefficient extension.

The system produces a full posterior distribution over team quality, enabling calibrated probabilistic statements about matchup outcomes. Ratings, predictions, and win probabilities all flow from the same generative object with quantifiable uncertainty at every level.

Reproducibility

All code is available in the accompanying repository. The dataset is derived from RealGM via an open scraper. The notebook `notebooks/probabilistic_ratings.ipynb` reproduces all figures and tables in this paper.

References

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